AWS Lab: Auto Scaling Group Golden AMI Scaling Under Load Triage

Duration : 60 Min.

Scenario

A web application managed by an Auto Scaling Group (ASG) is dropping client connections during a traffic surge because average CPU utilization is at 98%. The ASG is stuck at 2 instances (Min: 2, Max: 6, Desired: 2) because its CloudWatch scaling policy was misconfigured with an impossible metric threshold (CPU > 150%). The engineer must inspect the ASG configuration, correct the scaling policy to trigger scale-out when average CPU exceeds 70%, trigger the scale-out event, and verify that 2 new instances launch from the golden Launch Template.

Input Details

The cloud environment has been pre-configured with the following parameters: - **Platform:** AWS Auto Scaling Groups (ASG) & CloudWatch - **ASG Configuration:** `/opt/aws_asg/asg_config.json` - **Capacity Constraints:** Min: 2, Max: 6, Desired: 2 -> Scale out to 4 - **Target Metric Threshold:** Average CPU Utilization > 70%

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (derived from your login email before the `@` symbol, e.g. `student-demo01`). All custom resources and cloud configurations should incorporate this convention: - **Resource Naming:** Prepend or append `<prefix>` to your created AWS resources (e.g. `<prefix>-resource`). - **Candidate Tag:** Add tag `CandidatePrefix` = `<prefix>` on all managed cloud resources.

Task Objectives

Perform the following technical objectives directly in the AWS Management Console or using the AWS CLI: 1. Inspect ASG configuration and CloudWatch alarm in /opt/aws_asg/asg_config.json. 2. Identify the invalid CPU alarm threshold. 3. Update the target-tracking scaling policy to maintain 70% average CPU utilization. 4. Formulate the verification sequence showing ASG scaling from 2 to 4 instances and reaching InService state.

Verification

Once completed, run the verification script to check your progress and receive your score. The verification system will inspect the live resources in the AWS cloud:

```
python run.py
```

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Alarm defect identification: detect invalid metric threshold preventing automated scale-out	5 Marks
TC2	Policy correction: construct valid target tracking or step scaling policy with 70% threshold	5 Marks
TC3	Capacity scaling: verify DesiredCapacity increases from 2 to 4 instances	5 Marks
TC4	Health check verification: confirm new instances pass EC2 and ELB health checks	5 Marks

Total Score: 20 Marks

Important Notes

- Perform all actions directly in the AWS Management Console or using the AWS CLI.
- Ensure all resource configurations, tags, and naming conventions match the requirements.
- Do not terminate or alter pre-existing cloud infrastructure outside the scope of this assessment.